Humans Are More Diverse: Frontier LLMs Show Extreme Policies in Idealised AI Development Races

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Submission Id: TBD********†

ABSTRACT

An AI development race creates a multi-agent safety dilemma. Each company can develop slowly and safely, or move faster while taking a risk that may remove its final reward. We use this repeated game to study strategic safety behaviour among large language model (LLM) agents in races with two to five players. However, a valid action does not show that an agent understands the game. We therefore place an audit gate before behavioural interpretation. We first verify the game engine, then test rule recall, state tracking, payoff calculation, and stability under different but equivalent task descriptions. We then compare LLM action sequences with an evolutionary game-theory benchmark and published human data, and explore differences across models, risk conditions, personas, and two- to five-player races. The audit shows that strong rule recall can coexist with weak state tracking and expected-payoff calculation. Providing verified arithmetic and changing the response representation can also change later actions, even when the game rules stay fixed. Across seven tested model endpoints, aggregate rates hide large differences in action sequences, responses to opponents, and responses to race position. Patterns across the tested three- to five-player races are also model-specific rather than a single effect of adding competitors. These results show why multi-agent AI-race simulations need validity checks and trajectory-level analysis before their outputs are described as strategic, human-like, or safety-aware. Our findings are exploratory and apply only to the tested models, prompts, and decoding settings.

KEYWORDS

Large language models, AI development races, multi-agent systems, repeated games, AI safety, task validity

1 INTRODUCTION

Competition over advanced artificial intelligence can create a safety dilemma. Each company may benefit from moving faster, but safety work can slow it down when its rivals do not follow the same standard [5, 6, 10]. An *AI development race* is a simple game that represents this tension. In each round, every player chooses between slower Safe development and faster Unsafe development [11, 16]. The game is idealised: it does not model a real AI company in full. Its value is control. It lets us test how choices change with competition, relative progress, and the earlier actions of other players.

Human experiments show why the order of actions matters. In the two-player study of Domingos and Han [14], the preregistered comparison between the two higher private-risk conditions found no clear difference. Exploratory analyses instead linked later Unsafe choices to three parts of the race history: the opponent's previous action, whether the player was ahead or behind, and the player's first action. We call this *path dependence*: an early choice or event changes the later path of the interaction. A useful human and LLM comparison must therefore examine full action sequences, which we call *trajectories*, rather than only the overall Unsafe rate.

Large language models (LLMs) can act as language-based players in this game. They are already studied as simulated participants, decision-making agents, and game-playing systems [1, 2, 4]. Yet their use creates a measurement problem. A model may return a correctly formatted action without correctly tracking the state or calculating the payoff. In a repeated game, one such action changes the next state and can affect every later round. A plausible final trajectory is therefore not, by itself, evidence that the model understood the game.

Prompt-based evaluation is itself sensitive to presentation choices that do not necessarily change task meaning. Meaning-preserving few-shot formatting changes can produce large, model-dependent performance spreads [25]; reordering answer options can change multiple-choice predictions [22]; and order and response-token identity can create distinct selection biases [27]. Controlled studies also report answer changes under output-format requests and atomic whitespace variants [24]. Persona prompts should therefore be treated as experimental interventions rather than generic performance improvements: across a large factual-question evaluation, their average benefit was absent or slightly negative and the best persona was difficult to select reliably [31]. These findings motivate our separate audits of wording, answer order, arithmetic disclosure, narrative framing, and opaque response mapping.

Game-specific evaluations reveal the same validity threat in strategic choice. In Stag Hunt and Prisoner's Dilemma presentations, reversing the order of action labels and changing their payoff assignments altered choice frequencies, with model-specific positional and payoff biases [17]. Across procedurally generated Prisoner's Dilemma vignettes, decisions varied substantially with topic, actor, and world framing despite a common underlying game structure [23]. More generally, performance across 11 counterfactual tasks declined consistently when familiar default mappings were replaced by explicitly specified alternatives [28]. These results support our decision to test action-code mappings and narrative skins independently, and to avoid treating one familiar presentation as a neutral measurement instrument.

Reproducible game-based frameworks provide a complementary way to study these effects. FAIRGAME standardises repeated-game experiments across models, languages, and persona conditions, and supports comparison with game-theoretic predictions [8]. Later work extends this approach with a payoff-scaled Prisoner's Dilemma, a three-player Public Goods Game, and supervised recognition of canonical strategies from action trajectories [18]. A related payoff-scaling study finds that cooperation and inferred strategy distributions vary across models, languages, and incentive magnitudes, and can move in the opposite direction from its evolutionary benchmark [19]. These studies motivate our use of full trajectories and theoretical benchmarks, while our audit asks the separate question of whether an agent can correctly track the race that produces those trajectories.

Behavioural evaluations that focus on action frequencies or strategic outcomes can obscure this distinction unless task validity and representation robustness are established separately. An LLM may produce an aggregate UNSAFE rate similar to that of human participants while relying on incorrect state computations, prompt-specific heuristics, or unstable action encodings. Conversely, two checkpoints with similar aggregate action rates may exhibit substantially different responses to opponent behaviour, relative position, or early-round events. Aggregate behavioural similarity is therefore insufficient evidence that an LLM understands the task, reproduces human decision dynamics, or exhibits a general safety preference [7].

We address this problem with an *audit-first evaluation*: we check that the game and the agent's use of its rules are valid before we interpret the agent's behaviour. We reproduce the two-player game studied by Domingos and Han [14] and add a compatible multi-player version based on Han et al. [16]. All agents in a round see the same state before anyone acts. Their choices remain hidden until every agent has responded, so the choices are simultaneous. The game engine, not the LLM, then calculates progress, payoffs, the end of the race, prizes, ties, and setbacks. We save every prompt, response, parsed action, retry, state change, configuration, random seed, and final outcome.

The audit has four levels. First, *mechanical validity* means that the game engine applies the stated rules correctly. Second, *task validity* means that an agent can recall the rules, rebuild the current state, apply a state change, and calculate the final result. Third, *representation robustness* means that choices remain stable when wording or response labels change but the game itself does not. Fourth, we compare valid trajectories with published human patterns. The human data are a reference for observable behaviour. They are not a target that an LLM must copy, and LLM outputs are not treated as replacements for human participants.

That boundary is supported by mixed evidence from LLM-based behavioural simulation. Aher et al. reproduced qualitative findings in three of four experiments but found a hyper-accuracy distortion in the fourth [1]. Repeated-game work further shows that a model may predict an opponent's pattern without immediately acting in accordance with that prediction, while eliciting the prediction before action can improve coordination and score [2]. Accordingly, neither a plausible trajectory nor a verbal strategy report is treated here as sufficient evidence of task comprehension.

Distributional agreement is also stricter than matching a treatment mean. Dynamic market simulations have recovered broad differences between positive- and negative-feedback conditions while exhibiting less strategy heterogeneity than human participants [13]. Accordingly, our human comparison reports trajectory archetypes and within-population diversity alongside aggregate UNSAFE rates.

This emphasis on distributions rather than plausible prose is shared by broader evaluation work. CogBench phenotypes models with behavioural metrics from seven cognitive experiments [12], while SocioBench reports substantial individual- and subgroup-level gaps across large international survey data [26]. Persona-based simulation can reproduce linguistic patterns without recovering the full interaction of human perceptions [15], and specialised survey simulators still struggle on unseen questions [9]. Position papers consequently frame LLM social simulation as most defensible for pilot and exploratory use unless it is validated against the relevant population [3]. Strategic-game evidence is similarly heterogeneous: repeated Prisoner's Dilemma policies are not fully consistent across parameter and framing changes [21]; human-like heuristics can be applied more rigidly than by human participants [30]; multi-round agents may cooperate instead of converging to the predicted Nash equilibrium [29]; and real multi-turn action sequences remain difficult to reproduce from prompts alone [20].

We organise the paper around three questions about strategic behaviour in an AI race:

- **RQ1 - Strategic behaviour in two-player races:** How do agents respond to risk, opponent history, relative progress, and early choices, compared with game-theory and human benchmarks?
- **RQ2 - Multi-agent race dynamics:** Across races with three to five agents, how is Unsafe play associated with rank, model, persona, and tested group size?
- **RQ3 - Validity and strategic robustness:** Can agents track the game well enough for strategic interpretation, and are choices stable across equivalent task presentations?

The main insight is that the tested LLMs do not express one common strategic policy. In two-player races, models with similar aggregate Unsafe rates can respond differently to risk, opponent history, and relative progress. In three- to five-player races, the association between rank and Unsafe play can change with both model and persona, while aggregate behaviour does not move monotonically with group size. These are exploratory multi-agent patterns, not an identified causal effect of adding one more player. The validity audit also sets an important boundary: rule recall can coexist with incorrect state tracking, and an equivalent representation of the same game can change later actions. We reach that insight by connecting four views of one race: an auditable mechanism for two- and multi-player play, a comparison against game-theory and human benchmarks, an extension to three to five agents, and a validity gate applied before any strategic reading.

Each reported behavioural conclusion is scoped to its tested checkpoint, prompt version, decoding contract, experimental configuration, and run period. The labels SAFE and UNSAFE denote actions within the experimental game and should not be interpreted as direct measurements of general LLM safety, subjective risk preference, or suitability for autonomous deployment.

2 PRELIMINARIES

We first specify the two-player game reproduced from Domingos and Han [14] (§2.1), then its generalisation to $N > 2$ companies (§2.2). Every quantity that is not restated in §2.2, namely the horizon, prize split, and private-risk formula, carries over unchanged from the two-player game; only the stage payoff changes shape (Figure 1).

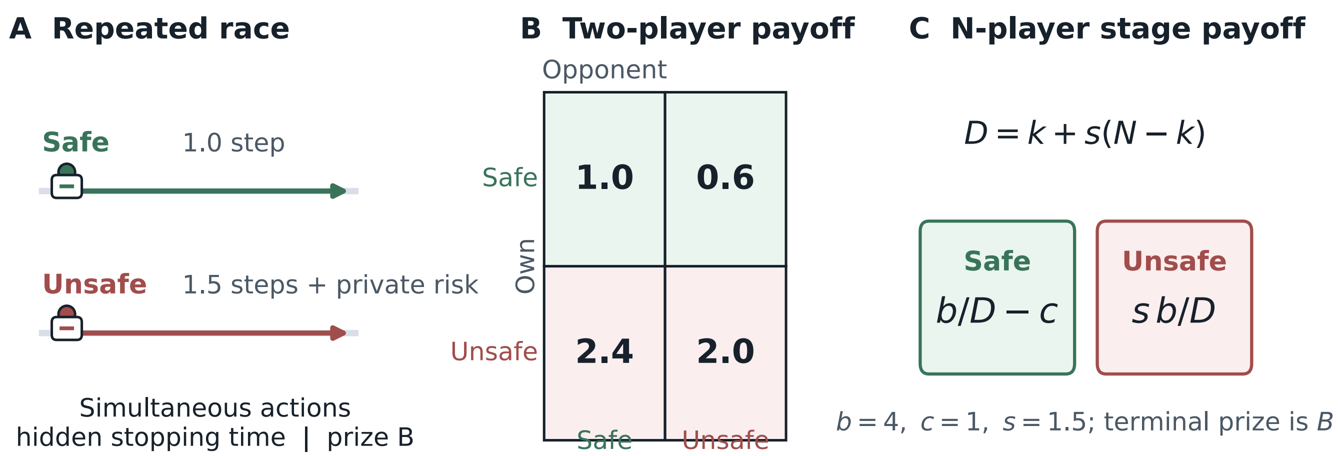


Figure 1: The AI-race mechanism, shared by the two-player and N-player games. (a) Each round every company advances by choosing Safe (green, slower, pays a cost) or Unsafe (red, faster, adding private setback risk); actions are simultaneous, the stopping time is hidden, and terminal prize B goes to the progress leader or is divided among tied leaders. (b) The two-player stage-payoff matrix, Equation 1. (c) The N-player stage-payoff rule, Equations 3 and 4, in which the per-round benefit b is distinct from terminal prize B.

2.1 Two-player game

Two companies $i \in \{1, 2\}$ play a repeated race. In every round t, both companies simultaneously choose an action $a_i^t \in \{S, U\}$ (Safe, Unsafe) and commit to it before either learns the other's choice for that round; only once both actions are committed does the round resolve (Figure 1a). Resolution has two parts. First, Safe advances the company's cumulative progress by $\sigma(S) = 1.0$ and Unsafe advances it by $\sigma(U) = 1.5$, so Unsafe always closes the race faster; progress accumulates as $P_i^t = P_i^{t-1} + \sigma(a_i^t)$. Second, the round pays a stage payoff $\pi_i^t = \pi(a_i^t, a_{-i}^t)$, read off the fixed matrix (own action by row, opponent action by column, Figure 1b):

$$\pi = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}. \tag{1}$$

Unsafe strictly dominates Safe at the stage-game level (row 2 exceeds row 1 in both columns): the only cost of Unsafe development enters later, through accumulated private risk rather than through the round payoff itself.

The race lasts a minimum of 5 rounds; after every completed round from round 5 onward it terminates with probability 0.2, giving an expected horizon $\mathbb{E}[T] = 9$. Neither company is ever told the horizon in advance, so a company can never condition its action on knowing the current round is the last one (Figure 1a). Termination then resolves in two further steps. First, the company with the higher cumulative progress P_i^T is declared the winner of a prize $B = 100$; if progress is exactly tied, the two companies split it 50/50. Second, and only for that winner or tied winners, an accumulated private setback risk is drawn:

$$q_i(T) = p_r^{\max} \cdot \frac{n_i^U(T)}{T}, q_i(T) \in [0, 1], p_r^{\max} \in \{0.10, 0.60, 0.90\}, \tag{2}$$

where $n_i^U(T)$ counts company i's Unsafe choices over the whole race and $p_r^{\max}$ is a treatment-level constant, assigned once per race, that we vary across three risk levels. If the draw against $q_i(T)$ fails, that company's entire race payoff, every stage payoff it accumulated plus the prize, is zeroed; if it succeeds, the company keeps both. A company that ends behind is never subject to this draw at all: it keeps its accumulated stage payoffs regardless of how much Unsafe it played, but receives no share of the prize (Figure 1a). This asymmetry, in which risk is built by both companies but realised only against whoever is ahead, is what makes Unsafe development a genuine gamble rather than a free speed advantage.

2.2 N-player game

The N-player game keeps every element of §2.1 unchanged (simultaneous sealed actions, progress increments $\sigma(S) = 1.0/\sigma(U) = 1.5$, the minimum-5-round hidden horizon, terminal prize $B = 100$, and the risk formula, Equation 2), but is played by $N \geq 2$ companies indexed by $i \in \{1, \ldots, N\}$ instead of exactly two, and the prize at round T splits evenly among however many companies share the highest progress rather than only two. The engine supports $N \in \{2, 3, 4, 5\}$. The two-player condition reproduces the original human-study mechanism, while the current multiplayer experiments cover matched conditions with $N \in \{3, 4, 5\}$ (Figure 1c).

With more than two companies there is no single opponent to index a matrix against, so Equation 1 generalises to the N-player stage-payoff rule of Han et al. [16] (their Appendix B), evaluated at $k^t \in \{0, 1, \ldots, N\}$, the number of companies (out of N) choosing Safe in round t. Therefore, every Safe-choosing company earns

$$\pi_{\text{Safe}}(k^t) = \frac{b}{k^t + s\,(N - k^t)} - c, \tag{3}$$

and every Unsafe-choosing company earns

$$\pi_{\text{Unsafe}}(k^t) = s \cdot \frac{b}{k^t + s\,(N - k^t)}, \tag{4}$$

with cost $c = 1$, per-round benefit $b = 4$, and speed $s = 1.5$ fixed across all group sizes (the per-round benefit b is distinct from terminal prize B, which is paid once, at the end of the race, not every round). Both payoffs depend on the *group's* joint action counts, k^t and $N - k^t$, not on any single rival's action, which is the structural difference from the two-player game: Thus, stage payoffs depend on the number of rivals choosing each action, not on their identities.

2.3 Game-theoretic benchmark

The game contains two opposing incentives. In a single round, Unsafe strictly dominates Safe: it gives more progress and a higher stage payoff against either action. Across the full race, however, Unsafe choices increase the setback risk for a winner or tied winner. The best action can therefore depend on the risk condition, earlier actions, and the expected response of other players. This is the strategic tension that the LLM agents must navigate.

We use four simple strategies as a theory benchmark: Always Safe (AS), Always Unsafe (AU), Conditional Safe (CS), which starts Safe and then copies the opponent, and Conditional Unsafe (CAS), which starts Unsafe and then copies the opponent. In the reconstructed finite-population evolutionary model of Domingos and Han [14], the most common strategy changes from AU at $p_r^{\max} = 0.10$, to CAS at 0.60, and to CS at 0.90. Thus, the theory does not predict one fixed amount of Unsafe play. It predicts a risk-dependent shift between unconditional speed, an aggressive opening, and conditional restraint. We use this as a qualitative benchmark, not as a fitted model of LLM output: prompted self-play trajectories are not samples from an evolutionary population.

3 EXPERIMENTAL DESIGN AND EVIDENCE POLICY

3.1 Agent protocol and units of analysis

At each decision, an agent receives a versioned prompt containing the current round, assigned maximum private risk, the publicly disclosed state, and the preceding revealed action profile when $t > 1$. Same-round decisions are sealed: every response is obtained from the same pre-action snapshot before the engine resolves actions, progress, payoffs, and risk. The environment, rather than generated text or arithmetic, determines all transitions. Raw response, parsed action, retry count, parser status, pre-turn state, terminal record, configuration, and seed allocation are retained for every decision.

The main behavioural outcome is whether an agent chooses Unsafe. The race, not the individual decision, is the main independent experimental unit. Decisions from the same race depend on earlier states and therefore cannot be treated as separate random samples. We report the number of races, trajectories, and decisions. Where possible, uncertainty is grouped by race or matched repetition. A *parse failure* occurs when a response does not follow the required action format. Because the resulting fallback action changes later states, one parse failure marks the whole race as contaminated.

3.2 Evidence strata and admission

We separate mechanical validity, task validity, diagnostic pilots, and confirmatory evidence. A *diagnostic pilot* is a small exploratory test used to find possible effects or failures. It does not estimate a stable population effect. A *confirmatory* study tests a question under a plan fixed before its results are inspected. A result enters that stronger evidence level only when the protocol, model identifier, prompt and configuration hashes, completed-race count, and exclusion rules were all fixed and recorded.

Table 1: Evidence strata used in this paper. Pilot modules are never pooled with confirmatory inference.

Module	Primary units	Observed outputs	Use
Frontier endpoint admission (Gemini 3 Flash, Claude Sonnet 5)	120 probes	120 answers	confirmatory gate
Calculator diagnostic	60 races	1,116 decisions	paired diagnostic
Context pilot ($T = 0$)	768 races	13,680 decisions	robustness diagnostic
Fixed-state replay	1,536 state cells	1,536 responses	direct prompt diagnostic
Five-checkpoint baseline	150 races	2,790 decisions	descriptive only
OpenAI $N = 3$ to 5 baseline	180 races	6,120 decisions	exploratory only

The primary audit uses protocol ai-race-game-understanding-v2: 41 atomic probes spanning rule recall, one-stage payoffs, state reconstruction, state transition, terminal scoring, and expected payoff. Numeric probes vary direct wording, paraphrase, and calculator disclosure; categorical probes also reverse answer order. Strict output-format compliance and semantic correctness are scored separately. A paired behavioural diagnostic compares the canonical prompt with a deterministic four-row decision card that discloses the focal agent's immediate payoff, progress, and private-risk consequences for each action profile. The card does not reveal the opponent's move or the stopping horizon.

We also vary the persona framing assigned to each company's executive across three families, shown in Figure 2: no persona at all, a risk-aware family with six levels, and a family that frames rivals as cooperative or adversarial. A neutral placebo of matched length separates the persona itself from the effect of adding a sentence.

The context extension keeps the game rules fixed but presents the game through eight different stories, which we call *narrative skins*. It also replaces Safe and Unsafe with the opaque response codes P and Q. We study this in three ways. A paired first-round comparison tests the same initial state. A *fixed-state replay* asks for a new action at a previously recorded state without allowing that answer to change later states. It therefore measures the direct effect of the prompt at that state. A *live trajectory* lets each answer change the next state, so direct prompt effects and later feedback can build on each other. A separate comprehension test acts as an admission gate. If that gate fails, a behavioural difference shows prompt-conditioned output, not informed optimisation of the game.

4 RESULTS

We report the validity gate first because every strategic claim depends on it. We then answer RQ1 by comparing two-player LLM trajectories with the evolutionary and human benchmarks, including responses to risk, opponent history, and relative progress. We answer RQ2 with the three- to five-player pilots, focusing on rank and persona. Representation diagnostics answer RQ3 and define where the behavioural interpretation must stop. The Appendix

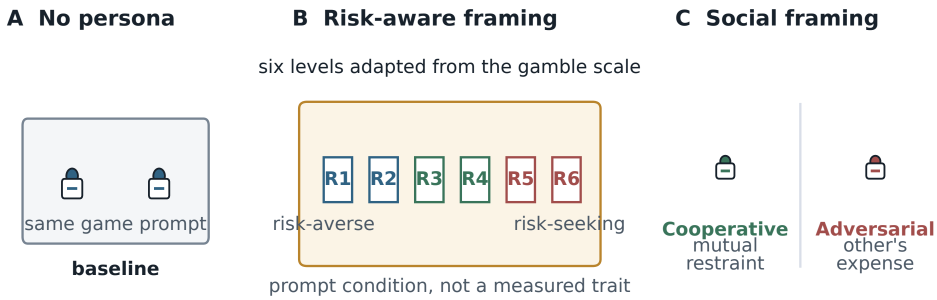


Figure 2: The three persona framings used to describe each company's executive. The canonical baseline has no persona sentence, and a neutral placebo of matched length separates the persona from its wording. The risk-aware family has six levels adapted from the Eckel and Grossman gamble scale. The cooperative and adversarial family frames rivals as doing better through mutual restraint, or only at each other's expense. These are prompt conditions, not measured psychological traits.

Table 2: Semantic accuracy on the 60-probe frontier endpoint admission for the two admitted routes, by probe subtask.

Subtask	Gemini 3 Flash	Claude Sonnet 5
Rule recall	100.0%	100.0%
One-stage payoff lookup	100.0%	100.0%
Terminal scoring	100.0%	80.0%
State reconstruction	93.3%	86.7%
State transition	100.0%	100.0%
Expected-payoff calculation	50.0%	0.0%
Overall (all subtasks pooled)	93.3%	81.7%

contains full model tables, predictive diagnostics, and robustness checks.

4.1 Validity gate: can agents follow the race?

We first ask whether the admitted frontier endpoints can carry out the rules they receive. The confirmatory endpoint admission contains 60 requested-seed outputs for each of two admitted frontier routes; all 120 planned outputs were retained and no transport or parse failure occurred. The hosted SDK did not apply the requested seed for Gemini and did not confirm provider application for Claude, as recorded in the manifests. As Table 2 shows, both routes reconstructed the tested state and terminal consequences above the gameplay threshold. They made errors on the closed-form expected-payoff probes, so those probes remain a diagnostic rather than a gameplay exclusion criterion. This separation prevents a failure on optional arithmetic from being mistaken for a failure to follow the action protocol.

The remaining gap is concentrated in closed-form expected-payoff arithmetic, with an additional Claude error on terminal scoring. We therefore retain the stricter per-endpoint audit and report the arithmetic diagnostic instead of treating frontier routes as interchangeable. Full probe-by-probe outputs, including failed endpoint admissions, are given in the supplementary material.

Table 3: Disclosed-arithmetic diagnostic: canonical vs. decision-card conditions (30 races, 558 decisions each, zero parse failures).

Condition	UNSAFE rate	95% CI	Mean payoff
Canonical	52.0%	48.0 to 55.9%	42.77
Decision card	60.8%	51.2 to 67.9%	42.21

4.2 Strategic robustness under equivalent presentations

We next ask whether sampled behaviour is stable under manipulations that leave the underlying game mechanics unchanged.

Disclosed arithmetic. The canonical and decision-card conditions each completed 30 races and 558 decisions with zero parse failures; matched risk-by-repetition cells had identical realised horizons. Table 3 compares the two conditions. Only 3.3% of paired first-round decisions changed, so most divergence appeared later in the race histories, after feedback had accumulated. These values show that supplying current-round arithmetic changes sampled behaviour; they do not by themselves identify an internal world model or a confirmatory causal effect.

Opaque symbolic mapping. The earlier local context-skin pilot is not part of the frontier evidence set and is excluded from the headline tables. Its mapping-by-repetition confound and failed comprehension gate are retained in the audit trail as a reason to require a fully crossed, per-endpoint frontier rerun. Until that rerun is complete, we make no causal claim about symbol mapping or narrative skins.

Together, these diagnostics show that action sequences can shift under changes that a correct game model should ignore. This is why the comprehension check in §4.1 comes before behavioural interpretation.

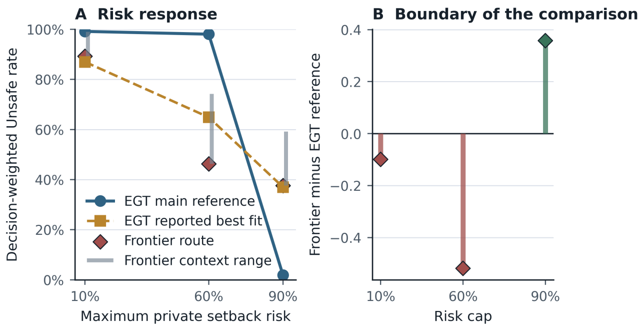
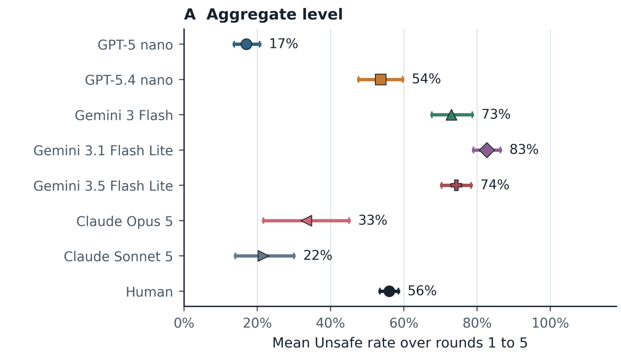


Figure 3: Frontier Unsafe decisions against the reconstructed evolutionary benchmark. The theory curves are stationary strategy mixtures; the frontier ranges and route-specific points summarise two admitted routes, each with 30 completed races. Independent-chain ranges are shown for the theory reconstruction.



Points are player-race trajectories; whiskers are trajectory bootstrap intervals.

Figure 4: Player-level mean UNSAFE rate over rounds 1 to 5 per population. Points are trajectory means and whiskers are bootstrap intervals over player-race trajectories.

4.3 Game theory versus prompted race behaviour

The evolutionary benchmark predicts a sharp change with private risk. At its main reference setting, predicted Unsafe play is 99.2%, 98.0%, and 1.9% in the low-, medium-, and high-risk conditions. This pattern follows the shift from AU to CAS and then CS described in §2.3. The frontier confirmatory baselines instead produce the following Unsafe rates at maximum risks 0.1, 0.6, and 0.9, respectively: Gemini 3 Flash, 98.9%, 74.2%, and 59.1%; Claude Sonnet 5, 89.2%, 46.2%, and 37.6%. Each route contributed 30 completed races and 558 decisions. These are descriptive endpoint results, not equilibrium estimates. The Gemini SDK route stripped the requested seed, while the Claude route forwarded it without provider-side application confirmation; both facts are retained in the run manifests.

This is a boundary rather than a failed equilibrium test. The evolutionary model describes strategy frequencies under selection and mutation; the LLM experiment samples prompted self-play decisions. Their units are different. The comparison nevertheless gives a useful game-theoretic insight: holding the payoff mechanism fixed does not guarantee the behavioural pattern predicted by the reduced strategy model. For the audited checkpoint, task representation can be more visible in observed play than the theoretical risk transition.

4.4 Two-player strategic diversity: humans and LLMs

We compare model trajectories with the human benchmark of Domingos and Han [14] directly on the literal first-five-round action-and-position histories, without first reducing them to hand-designed summary statistics. The comparison includes 420 trajectories from the seven tested models' two-player baseline (60 per model: GPT-5-nano, GPT-5.4-nano, Gemini-3-Flash, Gemini-3.1-Flash-Lite, Gemini-3.5-Flash-Lite, Claude Opus 5, and Claude Sonnet 5) and 340 complete human trajectories from the public de-identified dataset (one incomplete human record excluded); pooled $N = 760$. This seven-checkpoint comparison is a descriptive historical pilot: the current endpoint-admission artifacts cover Gemini 3 Flash and Claude Sonnet 5, so we do not use the other five checkpoints as evidence of cross-model task validity. Each player is represented by 15 raw features,

$$(\mathrm{own}_{1..5},\ \mathrm{opp}_{1..5},\ \mathrm{gap}_{1..5}),$$

own action, opponent action, and own-minus-opponent progress gap entering each of rounds 1 to 5, the recorded state and actions themselves, rather than derived quantities such as unsafe rate or retaliation.

Mean UNSAFE play over rounds 1 to 5 is lowest for GPT-5-nano (17%), followed by Claude Sonnet 5 (22%), Claude Opus 5 (33%), GPT-5.4-nano (54%), and humans (56%); the Gemini models range from 73% (Gemini-3-Flash) to 83% (Gemini-3.1-Flash-Lite). No single aggregate rate characterises the tested models, and closeness to the human mean is, on its own, uninformative about anything beyond that one number.

We next ask whether the tested checkpoints cover the diversity observed in humans. We fit k-means with $k = 4$ to five trajectory descriptors from the 341-person human sample only: overall UNSAFE rate, opponent reciprocity, position sensitivity, own-action persistence, and first-round UNSAFE play. The resulting groups are exploratory behavioural archetypes, not latent psychological types. Figure 5 shows their standardized signatures and the nearest-archetype projection of each model. The human groups are Persister ($n = 19$), Aggressive starter / reciprocator ($n = 170$), Cautious starter ($n = 133$), and Reciprocal catch-up ($n = 19$). GPT-5-nano maps almost entirely to Cautious starter, while GPT-5.4-nano is represented in all four groups. The Gemini checkpoints concentrate in the Aggressive starter / reciprocator and Reciprocal catch-up groups. These proportions describe coverage of a human reference space; they do not show that a model has recovered a human strategy or mechanism.

The embedding in Figure 6 makes the difference between behavioural *level* and behavioural *diversity* visible: GPT-5-nano is concentrated in the low-UNSAFE region, Gemini-3.1-Flash- Lite is concentrated near the high-UNSAFE region, and human trajectories

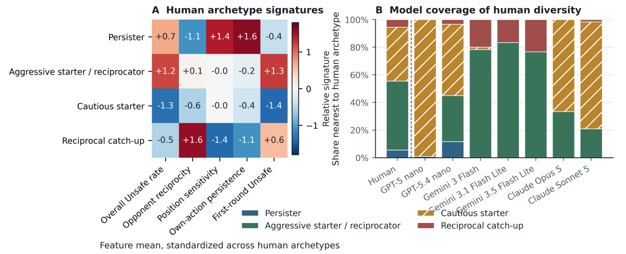


Figure 5: Human-reference behavioural archetypes and model coverage. Panel A shows the five descriptor means for the four human-only k-means groups, standardized across groups. Panel B assigns each model trajectory to its nearest human archetype after human-based standardization. The dashed line separates the human reference from the model projections.

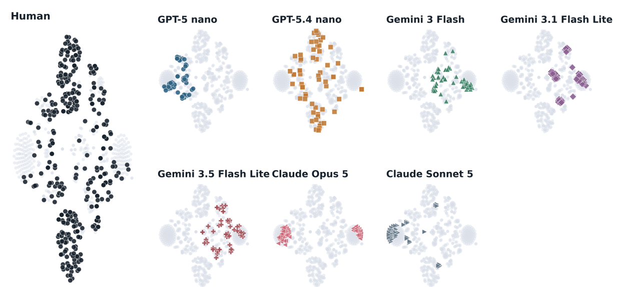


Figure 6: t-SNE visualisation of the pooled 15-dimensional raw action-and-position space. Each panel highlights one population while rendering the other populations in grey; the human sample is shown in the larger left panel, with the seven LLM checkpoints arranged smaller to its right.

occupy a much broader portion of the observed space, overlapping several model-specific regions.

Recolouring the identical coordinates by outcome instead of identity, shown in the supplementary material, separates the embedding almost entirely by UNSAFE rate: a solid-green region, a solid-red region, and a mixed band in between. The archetype plot and this outcome-coloured embedding answer different questions. The former summarizes five interpretable descriptors; the latter shows geometry in the raw sequence space. Neither should be read as a definitive clustering of latent strategies.

We next test whether five rounds contain enough information to identify the population that produced a trajectory. A small decision tree reaches 38.9% ± 3.3 accuracy under five-fold cross-validation. Random guessing among eight balanced classes would give 12.5%. Population identity is therefore visible, but the groups still overlap. The tree correctly identifies 96.7% of Gemini-3.1-Flash-Lite trajectories. For GPT-5-nano, this value is 63.3%, and 31.7% are instead labelled Claude Opus 5 because both often start with mutual Safe play. Only 33.2% of human trajectories are labelled human. This is the lowest value among the populations. Human actions overlap with several model patterns because the human sample is internally diverse, not because it follows one middle policy.

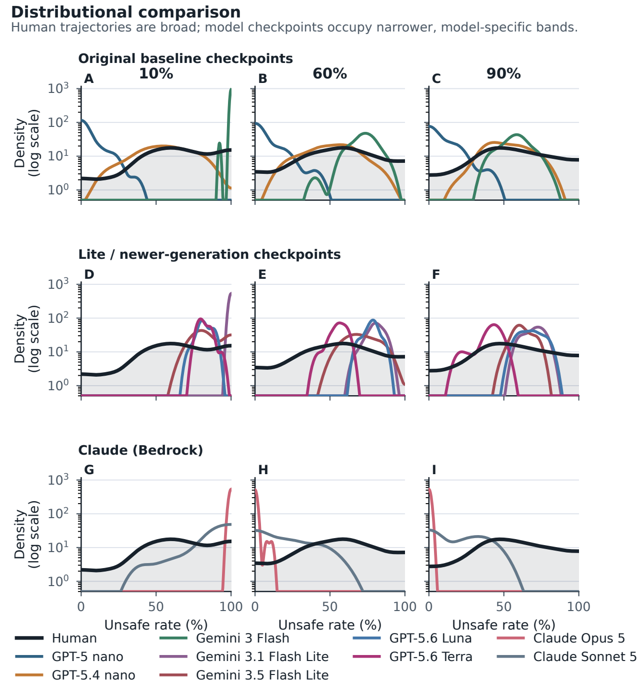


Figure 7: Smoothed distribution of player-level UNSAFE rates by risk cap (columns) and model family (rows), with the human benchmark repeated in every panel as the shaded reference. Density is log-scaled: model peaks span close to three orders of magnitude, and a linear axis would flatten all but the tallest. The middle row includes GPT-5.6 Luna and Terra, outside the seven-model roster.

The supplementary round-by-round profile gives a complementary view of the aggregate rate: the human profile lies close to the pooled eight-population mean on nearly every own-, opponent-, and gap-action axis, whereas GPT-5-nano and Claude Sonnet 5 are displaced toward SAFE, the two Gemini Flash-Lite checkpoints are displaced toward UNSAFE, and GPT-5.4-nano, Claude Opus 5, and Gemini-3-Flash fall between these extremes. Humans therefore do not occupy a single isolated behavioural corner: their aggregate profile is comparatively central, while their individual trajectories span the regions occupied by several different models.

This heterogeneity is not merely a visual impression. Fitting three nested logistic models to the pooled neutral-baseline pilot (risk-only, LLM-identity-only, and LLM-identity-by-risk interaction) shows that allowing the risk-response profile to vary by LLM improves fit substantially beyond LLM-specific intercepts alone,

$$\chi^2(10) = 354.7, \qquad p \approx 4 \times 10^{-70},$$

with several risk-by-LLM cells approaching complete separation (a convergence warning in the interaction model means the exact statistic should be read as approximate). The fit covers the five-model neutral baseline only; the two Claude models are not in it. Neither caveat changes the conclusion: no single risk-response curve represents all five, so cross-model differences are differences in *shape*, not only in level.

This distributional picture is visible directly in the raw UNSAFE-rate data as well (Figure 7): human participant-level UNSAFE rates spread across nearly the entire available range at each risk level, whereas each tested model occupies a comparatively narrow, model-specific band: GPT-5-nano concentrated near the low-UNSAFE end, the Gemini models toward the high-UNSAFE end, and GPT-5.4-nano in a more intermediate, diffuse region. Consequently, similarity between one model's aggregate UNSAFE rate and the human mean does not imply that the model reproduces the underlying human distribution.

A more limited correspondence appears in the relationship between race outcome and behaviour: human winners play UNSAFE 16.0 percentage points more often than human losers, versus an approximately 20-point pooled descriptive difference among the tested models. This agreement is confined to one marginal association and does not extend to the full dynamic structure: the human estimates show a precise positive association with the opponent's previous action and a negative association with relative race position, whereas several model estimates are weak, reversed in sign, or not reliably estimable because their action distributions sit near floor or ceiling. Differences in sign, magnitude, and precision across models confirm that agreement in an aggregate action rate does not imply agreement in the underlying behavioural dynamics.

4.5 Observable drivers of UNSAFE play

Aggregate action rates do not show how agents respond to a changing race. We therefore test how five pre-decision variables predict Unsafe play: the player's previous action, the opponent's previous action, progress gap, assigned private risk, and round number. These are predictive associations, not causal effects or evidence about internal reasoning. The full random- forest and SHAP analysis is reported in the supplementary material.

The human data are organised mainly by the opponent's previous action, which accounts for 48% of the fitted model's total SHAP magnitude. The tested LLMs do not share one pattern. Progress gap is largest for GPT-5-nano (44%), the assigned risk condition is largest for two Gemini checkpoints (35 to 40%), and opponent history is largest for Claude Sonnet 5 (51%). GPT-5.4-nano has no single strong predictor. These results support a game-theoretic point: two populations can have similar Unsafe rates while following different response rules. the supplementary material reports model fit, confounding, and floor/ceiling limits.

4.6 Multi-agent dynamics: position in N-player races

The two-player analysis uses the opponent's previous action and the exact progress gap. In races with more players, we use a simpler question: does an agent act differently when it is a Leader, in the Middle, or a Trailer? We compare these positions within four persona bands. Baseline adds no persona sentence; Low combines R1 to R2; Mid combines R3 to R4; and High combines R5 to R6. A Leader has the highest progress or is tied for it before the current round. A Trailer is strictly last. Middle covers every other position. These labels describe observed states; the experiment did not randomly assign rank.

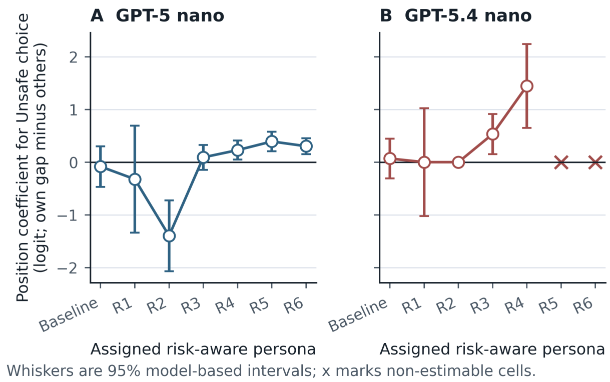


Figure 8: Position association in the N-player pilot. Each panel shows the model-based coefficient for the player's relative position within an assigned risk-aware persona, with the other players' position held in the contrast. Whiskers are 95% intervals. Crosses mark cells that were not estimable because the observed actions reached a floor or ceiling.

For GPT-5-nano, the position coefficient is negative at Baseline, R1, and R2, then positive from R3 through R6. This sign change is consistent with the qualitative "falling behind" account of Domingos and Han [14], but the coefficient is an observational association, not a causal effect of rank. Position is measured immediately before the action, so a player's previous actions help determine both its current rank and its next choice. The sign reversal can therefore arise from selection into ranks rather than from a change in the response to being ahead.

GPT-5.4-nano has coefficients close to zero through R2 and positive coefficients at R3 and R4. R5 and R6 are not estimable because the observed actions are at a floor or ceiling in those cells. The contrast with GPT-5-nano shows why the result should not be compressed into a single position effect: the association depends on the checkpoint and on the prompt-assigned persona. Separating persona from rank would require a design that crosses the two factors while controlling or randomizing prior action history.

Read together, the two models show that "does position matter" has no single answer even within one model family. The assigned persona is the more stable source of variation in this pilot, while position is a secondary association whose sign and estimability depend on the checkpoint. The bands here are the prompt-embedded risk *persona* examined next in §4.7, not the mechanism's private-risk parameter $p_r^{\max}$ (Eq. 2), and each band pools over $p_r^{\max}$ rather than fixing it. The distinction matters: the persona is an instruction re-assigned per race, $p_r^{\max}$ a property of the game. This section therefore shows rank associations modulated by persona; separating that from the mechanical risk treatment needs a design crossing the two, which this pilot lacks. The broader $N \in \{3, 4, 5\}$ self-play scope (180 pilot races, 6,120 decisions; aggregate UNSAFE rates of 16.2%, 26.8%, 21.9% for GPT-5-nano and 83.7%, 87.0%, 84.0% for GPT-5.4-nano at $N = 3, 4, 5$) is reported in the supplementary material; because changing N also changes the stage-payoff table, opponent count, and prompt length, we do not treat that comparison as an isolated group-size effect.

4.7 Measured human risk and prompted LLM personas

Human risk preference and an LLM risk persona are different variables. Human participants completed an incentivised risk task before the race. Their score has almost no direct relation with later Unsafe play ($r = -0.015$, $p = 0.79$, $n = 341$). By contrast, an LLM persona is an instruction placed inside every race prompt. Across models with a six-level sweep, moving from the lowest to the highest risk persona raises Unsafe play by 50.5 to 98.3 percentage points.

The contrast supports a behavioural, not psychological, interpretation. The human measure describes a person before play; the LLM label actively changes the decision context. A risk persona therefore acts as a strong policy prompt, not as evidence that the model has a stable risk preference. Full curves, model-level estimates, and scope limits are reported in the supplementary material.

5 DISCUSSION AND LIMITATIONS

The main lesson is not that an audited model "understands the game." The evidence supports a narrower claim. The admitted frontier endpoints can recall rules, reconstruct tested states, and apply terminal eligibility, yet still make errors on closed-form expected payoff. Reports of LLM behaviour in repeated games should therefore show per-endpoint comprehension accuracy and parser health together with action rates.

The human experiment gives us an external behavioural reference. It is not a target that an LLM should reproduce. When an LLM matches a human association, the two populations behave similarly on that measured part of this task. This does not show that they use the same reasoning or psychology. A mismatch also does not make the human result invalid. Model, persona, and multiplayer comparisons have the same boundary: each applies only to the tested endpoint, prompt, decoding setting, provider route, and run period.

Several limitations shape what we can claim. First, the earlier context-to-symbol mapping changes with repetition parity, so mapping and repetition are not fully separated. Its comprehension gate also failed. We therefore exclude this pilot from headline evidence and require a fully crossed, per-endpoint frontier rerun before making a mapping claim. The two current frontier endpoint admissions pass the state and terminal gates but miss some closed-form expected-payoff probes; this is reported as a diagnostic limitation, not hidden by a fallback parser. The seven-checkpoint diversity analysis remains descriptive because the other five checkpoints lack the same endpoint-admission artifact. Fixed-state replay measures one decision per saved state, whereas live play measures dependent decisions along a changing trajectory, so their effect sizes are not interchangeable. Multiplayer pilots are small and do not include a matched two-player condition. Changing player count also changes the payoff table, opponent count, and prompt length, so we cannot identify a pure group-size effect. Frontier endpoint availability and seed forwarding are provider-specific: the current confirmatory route requested seeds but the hosted SDK stripped them, and only one authorised Kaggle identity was configured. The EGT comparison is a repo-native reconstruction with an EGTtools source audit, not bitwise reproduction of undisclosed author code. More broadly, this game models only a private speed and safety trade-off; it does not model the full structure of frontier AI labs, collective harm, regulation, communication, or uncertainty about real capabilities.

6 REPRODUCIBILITY AND RESPONSIBLE REPORTING

Each admitted run is represented by append-only event logs and a manifest recording the model route, prompt and source hashes, decoding contract, seed requests, parser status, and execution metadata. Derived analyses are rebuilt from event logs rather than console output, and failed sessions or exclusions remain in the accounting trail. Pilot and confirmatory evidence are not pooled. We avoid anthropomorphic interpretations of actions and release code, prompts, manifests, and de-identified logs subject to provider, licence, and privacy constraints. The canonical frontier outputs are kept in the repository's `paper/` and `results/frontier/` trees; the submission PDFs are built into `paper/` by the repository build workflow.

7 CONCLUSION

Strategic safety behaviour in these AI races is not a fixed property of an LLM. In two-player races, models differ in their responses to risk, opponents, and relative progress. Across three to five agents, position patterns vary by model and persona and are not monotone in group size. Yet rule recall does not ensure state tracking, and equivalent presentations can change later play. Multi-agent AI-race studies should therefore report dynamic response rules and validity, not only aggregate Unsafe rates. These findings remain exploratory until matched confirmatory runs pass the same gates.

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